

Basics of Robotics

□ Learn the types of robots, their anatomy, and key components like actuators and sensors

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 6492B is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 6492B.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Microelectronics
Course code	6492B
Course title	Basics of Robotics
Semester	6 Credits: 4
Course category	Open Elective
Periods per week	4 (L:4 T:0 P:0) Periods per semester: 60

Item	Official information
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

17 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- Learn the types of robots, their anatomy, and key components like actuators and sensors
- Analyze factors such as axes, precision, and capacity to choose appropriate robots for various applications.
- Understand different sensor and actuator types and how to select them for specific robotic
- Learn basic kinematics principles and use transformation matrices for solving forward and inverse kinematics problems.

Course outcomes

CO	Official outcome	Study level
CO1	applications of Robots 14 Remembering Select suitable sensors and actuators for robotic	See official syllabus
CO2	14 Understanding systems. Identify the right robotic configuration and gripper	See official syllabus
CO3	14 Understanding for specific tasks. Determine the most suitable robotic configuration	See official syllabus
CO4	and gripper for a specific application. 16 Applying	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3 2
CO2	CO2 3 3 3 2 2
CO3	CO3 3 3 3 2 3

Row	Extracted official mapping
CO4	CO4 3 3 3 3 3 1

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Familiarize with anatomy, specifications and applications of Robots	4	As specified
Module 2	Select suitable sensors and actuators for robotic systems.	4	As specified
Module 3	Identify the right robotic configuration and gripper for specific tasks.	4	As specified
Module 4	specific application.	5	As specified

MODULE 1

Familiarize with anatomy, specifications and applications of Robots

This module is presented from the official syllabus scope for Basics of Robotics. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.

- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Familiarize with anatomy, specifications and applications of Robots
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

3 Remembering types of robots (manipulators, mobile, aerial). Describe the anatomy of a robotic manipulator,

3 Remembering types of robots (manipulators, mobile, aerial). Describe the anatomy of a robotic manipulator, is an official topic in Module 1 of Basics of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Remembering types of robots (manipulators, mobile, aerial). Describe the anatomy of a robotic manipulator, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 remembering types of robots (manipulators, mobile, aerial). describe the anatomy of a robotic manipulator, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Remembering types of robots (manipulators, mobile, aerial). Describe the anatomy of a robotic manipulator, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-3 Remembering types of robots (manipulators, mobile, aerial). Describe the anatomy of a robotic manipulator, $\square\square\square\square\square\square\square\square\square$ $\square\square\square\square$ definition/meaning $\square\square\square\square\square\square\square\square\square$. $\square\square\square\square$ $\square\square\square\square$ $\square\square\square\square$, steps, application $\square\square\square\square$ $\square\square\square\square\square\square\square$ $\square\square\square\square$. Technical terms English- $\square$ $\square\square\square\square$ $\square\square\square\square\square\square\square$.

including links, joints, actuators, and 5 Understanding controllers. Explain kinematic chains, degrees of freedom,

including links, joints, actuators, and 5 Understanding controllers. Explain kinematic chains, degrees of freedom, is an official topic in Module 1 of Basics of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify including links, joints, actuators, and 5 Understanding controllers. Explain kinematic chains, degrees of freedom, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, including links, joints, actuators, and 5 understanding controllers. explain kinematic chains, degrees of freedom, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What including links, joints, actuators, and 5 Understanding controllers. Explain kinematic chains, degrees of freedom, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-ഇംഗ്ലീഷ് ഇൻക്ലൂഡിംഗ് ലിങ്ക്സ്, ജോയിന്റ്സ്, ആക്ടുേറ്റേഴ്സ്, and 5 Understanding controllers. Explain kinematic chains, degrees of freedom, ഡിഗ്രീസ് ഓഫ് ഫ്രീഡം definition/meaning ഡിഗ്രീസ് ഓഫ് ഫ്രീഡം. ഡിഗ്രീസ് ഓഫ് ഫ്രീഡം steps, application ഡിഗ്രീസ് ഓഫ് ഫ്രീഡം ഡിഗ്രീസ് ഓഫ് ഫ്രീഡം. Technical terms English-ഇംഗ്ലീഷ് ഡിഗ്രീസ് ഓഫ് ഫ്രീഡം.

4 Understanding and operational considerations for robots. List applications of robots in various fields like

4 Understanding and operational considerations for robots. List applications of robots in various fields like is an official topic in Module 1 of Basics of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding and operational considerations for robots. List applications of robots in various fields like using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding and operational considerations for robots. list applications of robots in various fields like should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding and operational considerations for robots. List applications of robots in various fields like means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-4 Understanding and operational considerations for robots. List applications of robots in various fields like $\text{പ്രകൃതിദൃഷ്ടിയിൽ പ്രവർത്തിക്കുന്ന റോബോട്ടുകൾ}$ definition/meaning $\text{പ്രകൃതിദൃഷ്ടിയിൽ പ്രവർത്തിക്കുന്ന റോബോട്ടുകൾ}$. $\text{പ്രകൃതിദൃഷ്ടിയിൽ പ്രവർത്തിക്കുന്ന റോബോട്ടുകൾ}$, steps, application $\text{പ്രകൃതിദൃഷ്ടിയിൽ പ്രവർത്തിക്കുന്ന റോബോട്ടുകൾ}$. Technical terms English- $\text{പ്രകൃതിദൃഷ്ടിയിൽ പ്രവർത്തിക്കുന്ന റോബോട്ടുകൾ}$.

2 Remembering medical, industrial, mining, and space. Definition, Anatomy, and Applications of Robots Definitions- Robots, Robotics; Types of Robots- Manipulators, Mobile Robots-wheeled & Legged Robots, Aerial Robots; Anatomy of a robotic manipulator-links, joints, actuators, sensors, controller; open kinematic vs closed kinematic chain; degrees of freedom; Robot considerations for an application- number of axes, work volume, capacity & speed, stroke & reach, Repeatability, Precision and Accuracy, Operating environment, point to point control or continuous path control., Robot Applications- medical, mining, space, defense, security, domestic, entertainment, Industrial Applications-Material handling, welding, Spray painting, Machining.

2 Remembering medical, industrial, mining, and space. Definition, Anatomy, and Applications of Robots Definitions- Robots, Robotics; Types of Robots- Manipulators, Mobile Robots-wheeled & Legged Robots, Aerial Robots; Anatomy of a robotic manipulator-links, joints, actuators, sensors, controller; open kinematic vs closed kinematic chain; degrees of freedom; Robot considerations for an application- number of axes, work volume, capacity & speed, stroke & reach, Repeatability, Precision and Accuracy, Operating environment, point to point control or continuous path control., Robot Applications- medical, mining, space, defense, security, domestic, entertainment, Industrial Applications- Material handling, welding, Spray painting, Machining. is an official topic in Module 1 of Basics of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Remembering medical, industrial, mining, and space. Definition, Anatomy, and Applications of Robots Definitions- Robots, Robotics; Types of Robots- Manipulators, Mobile Robots-wheeled & Legged Robots, Aerial Robots; Anatomy of a robotic manipulator-links, joints, actuators, sensors, controller; open kinematic vs closed kinematic chain; degrees of freedom; Robot considerations for an application- number of axes, work volume, capacity & speed, stroke & reach, Repeatability, Precision and Accuracy, Operating environment, point to point control or continuous path control., Robot Applications- medical, mining,

space, defense, security, domestic, entertainment, Industrial Applications-Material handling, welding, Spray painting, Machining. using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 remembering medical, industrial, mining, and space. definition, anatomy, and applications of robots definitions- robots, robotics; types of robots- manipulators, mobile robots-wheeled & legged robots, aerial robots; anatomy of a robotic manipulator-links, joints, actuators, sensors, controller; open kinematic vs closed kinematic chain; degrees of freedom; robot considerations for an application- number of axes, work volume, capacity & speed, stroke & reach, repeatability, precision and accuracy, operating environment, point to point control or continuous path control., robot applications- medical, mining, space, defense, security, domestic, entertainment, industrial applications-material handling, welding, spray painting, machining. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Remembering medical, industrial, mining, and space. Definition, Anatomy, and Applications of Robots Definitions- Robots, Robotics; Types of Robots- Manipulators, Mobile Robots- wheeled & Legged Robots, Aerial Robots; Anatomy of a robotic manipulator-links, joints, actuators, sensors, controller; open kinematic vs closed kinematic chain; degrees of freedom; Robot considerations for an application- number of axes, work volume, capacity & speed, stroke & reach, Repeatability, Precision and Accuracy, Operating environment, point to point control or continuous path control., Robot Applications- medical, mining, space, defense, security, domestic, entertainment, Industrial Applications-Material handling, welding, Spray painting, Machining. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- 2 Remembering medical, industrial, mining, and space. Definition, Anatomy, and Applications of Robots Definitions- Robots, Robotics; Types of Robots- Manipulators, Mobile Robots-wheeled & Legged Robots, Aerial Robots; Anatomy of a robotic manipulator-links, joints, actuators, sensors, controller; open kinematic vs closed kinematic chain; degrees of freedom; Robot considerations for an application- number of axes, work volume, capacity & speed, stroke & reach, Repeatability, Precision and Accuracy, Operating environment, point to point control or continuous path control., Robot Applications- medical, mining, space, defense, security, domestic, entertainment, Industrial Applications-Material handling, welding, Spray painting, Machining.
 definition/meaning
 steps, application
 Technical terms English-

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

Familiarize with anatomy, specifications and applications of Robots

M1.01 Remembering	Recall the definitions of robots, robotics, and types of robots (manipulators, mobile, aerial).	3
M1.02 Understanding	Describe the anatomy of a robotic manipulator, including links, joints, actuators, and controllers.	5
M1.03 Understanding	Explain kinematic chains, degrees of freedom, and operational considerations for robots.	4
M1.04 Remembering	List applications of robots in various fields like medical, industrial, mining, and space.	2

Contents:
Definition, Anatomy, and Applications of Robots
Definitions- Robots, Robotics; Types of Robots- Manipulators, Mobile Robots- wheeled & Legged Robots, Aerial Robots; Anatomy of a robotic manipulator-links, joints, actuators, sensors, controller; open kinematic vs closed kinematic chain; degrees of

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Select suitable sensors and actuators for robotic systems.

This module is presented from the official syllabus scope for Basics of Robotics. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Select suitable sensors and actuators for robotic systems.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

robotics, such as touch, force, proximity, and 3 Remembering vision sensors. Explain the difference between internal and

robotics, such as touch, force, proximity, and 3 Remembering vision sensors. Explain the difference between internal and is an official topic in Module 2 of Basics of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify robotics, such as touch, force, proximity, and 3 Remembering vision sensors. Explain the difference between internal and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, robotics, such as touch, force, proximity, and 3 remembering vision sensors. explain the difference between internal and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What robotics, such as touch, force, proximity, and 3 Remembering vision sensors. Explain the difference between internal and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-റൂട്ടിംഗ് robotics, such as touch, force, proximity, and 3 Remembering vision sensors. Explain the difference between internal and ഫോട്ടോറീഫ്ലക്സ് ഫോട്ടോറീഫ്ലക്സ് definition/meaning ഫോട്ടോറീഫ്ലക്സ്. ഫോട്ടോറീഫ്ലക്സ് ഫോട്ടോറീഫ്ലക്സ് ഫോട്ടോറീഫ്ലക്സ്, steps, application ഫോട്ടോറീഫ്ലക്സ് ഫോട്ടോറീഫ്ലക്സ് ഫോട്ടോറീഫ്ലക്സ്. Technical terms English-റൂട്ടിംഗ് ഫോട്ടോറീഫ്ലക്സ് ഫോട്ടോറീഫ്ലക്സ്.

external sensors, and describe vision sensor 4 Understanding elements. Describe the classification, working, and

external sensors, and describe vision sensor 4 Understanding elements. Describe the classification, working, and is an official topic in Module 2 of Basics of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify external sensors, and describe vision sensor 4 Understanding elements. Describe the classification, working, and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, external sensors, and describe vision sensor 4 understanding elements. describe the classification, working, and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What external sensors, and describe vision sensor 4 Understanding elements. Describe the classification, working, and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-പ്രാദേശിക സെൻസറുകൾ, and describe vision sensor 4 Understanding elements. Describe the classification, working, and പ്രവർത്തനം പ്രതിപാദിക്കുക definition/meaning പ്രതിപാദിക്കുക. പ്രവർത്തനം പ്രതിപാദിക്കുക, steps, application പ്രവർത്തനം പ്രതിപാദിക്കുക പ്രതിപാദിക്കുക. Technical terms English-ൽ പ്രതിപാദിക്കുക പ്രതിപാദിക്കുക.

advantages/disadvantages of electric, 4 Understanding hydraulic, and pneumatic actuators. Apply knowledge to select suitable sensors and

advantages/disadvantages of electric, 4 Understanding hydraulic, and pneumatic actuators. Apply knowledge to select suitable sensors and is an official topic in Module 2 of Basics of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify advantages/disadvantages of electric, 4 Understanding hydraulic, and pneumatic actuators. Apply knowledge to select suitable sensors and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, advantages/disadvantages of electric, 4 understanding hydraulic, and pneumatic actuators. apply knowledge to select suitable sensors and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What advantages/disadvantages of electric, 4 Understanding hydraulic, and pneumatic actuators. Apply knowledge to select suitable sensors and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a

diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-ഇന്റർഫേസ് ആവശ്യകതകൾ/ഇന്റർഫേസ് ആവശ്യകതകൾ, 4 Understanding hydraulic, and pneumatic actuators. Apply knowledge to select suitable sensors and ഇന്റർഫേസ് ആവശ്യകതകൾ definition/meaning ഇന്റർഫേസ് ആവശ്യകതകൾ. ഇന്റർഫേസ് ആവശ്യകതകൾ, steps, application ഇന്റർഫേസ് ആവശ്യകതകൾ ഇന്റർഫേസ് ആവശ്യകതകൾ. Technical terms English-ഇന്റർഫേസ് ആവശ്യകതകൾ.

3 Applying actuators for robotic applications. Sensors and Actuators Sensor classification- touch, force, proximity, vision sensors. Internal sensors- Position sensors, velocity sensors, acceleration sensors, Force sensors; External sensors-contact type, noncontact type; Vision - Elements of vision sensor, image acquisition, image processing; Selection of sensors. Actuators for robots- classification-Electric, Hydraulic, Pneumatic actuators; their advantages and disadvantages; Electric actuators- Stepper motors, DC motors, DC servo motors and their drivers, AC motors, Linear actuators, selection of motors; Hydraulic actuators- Components and typical circuit, advantages and disadvantages; Pneumatic Actuators- Components and typical circuit, advantages and disadvantages.

3 Applying actuators for robotic applications. Sensors and Actuators Sensor classification- touch, force, proximity, vision sensors. Internal sensors-Position sensors, velocity sensors, acceleration sensors, Force sensors; External sensors-contact type, noncontact type; Vision - Elements of vision sensor, image acquisition, image processing; Selection of sensors. Actuators for robots-classification-Electric, Hydraulic, Pneumatic actuators; their advantages and disadvantages; Electric actuators- Stepper motors, DC motors, DC servo motors and their drivers, AC motors, Linear actuators, selection of motors; Hydraulic actuators- Components and typical circuit, advantages and disadvantages; Pneumatic Actuators- Components and typical circuit, advantages and disadvantages. is an official topic in Module 2 of Basics of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Applying actuators for robotic applications. Sensors and Actuators Sensor classification- touch, force, proximity, vision sensors. Internal sensors-Position sensors, velocity sensors, acceleration sensors, Force sensors; External sensors-contact type, noncontact type; Vision - Elements of vision sensor, image acquisition, image processing; Selection of sensors. Actuators for robots-classification-Electric, Hydraulic, Pneumatic actuators; their advantages and disadvantages; Electric actuators- Stepper motors, DC motors, DC servo motors and their drivers, AC motors, Linear actuators, selection of motors; Hydraulic

actuators- Components and typical circuit, advantages and disadvantages;
Pneumatic Actuators- Components and typical circuit, advantages and disadvantages. using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 applying actuators for robotic applications. sensors and actuators sensor classification- touch, force, proximity, vision sensors. internal sensors- position sensors, velocity sensors, acceleration sensors, force sensors; external sensors- contact type, noncontact type; vision - elements of vision sensor, image acquisition, image processing; selection of sensors. actuators for robots- classification-electric, hydraulic, pneumatic actuators; their advantages and disadvantages; electric actuators- stepper motors, dc motors, dc servo motors and their drivers, ac motors, linear actuators, selection of motors; hydraulic actuators- components and typical circuit, advantages and disadvantages; pneumatic actuators- components and typical circuit, advantages and disadvantages. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Applying actuators for robotic applications. Sensors and Actuators Sensor classification- touch, force, proximity, vision sensors. Internal sensors-Position sensors, velocity sensors, acceleration sensors, Force sensors; External sensors-contact type, noncontact type; Vision - Elements of vision sensor, image acquisition, image processing; Selection of sensors. Actuators for robots- classification-Electric, Hydraulic, Pneumatic actuators; their advantages and disadvantages; Electric actuators- Stepper motors, DC motors, DC servo motors and their drivers, AC motors, Linear actuators, selection of motors; Hydraulic actuators- Components and typical circuit, advantages and disadvantages; Pneumatic Actuators- Components and typical circuit, advantages and disadvantages. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- 3 Applying actuators for robotic applications. Sensors and Actuators Sensor classification- touch, force, proximity, vision sensors. Internal sensors-Position sensors, velocity sensors, acceleration sensors, Force sensors; External sensors-contact type, noncontact type; Vision - Elements of vision sensor, image acquisition, image processing; Selection of sensors. Actuators for robots- classification-Electric, Hydraulic, Pneumatic actuators; their advantages and disadvantages; Electric actuators- Stepper motors, DC motors, DC servo motors and their drivers, AC motors, Linear actuators, selection of motors; Hydraulic actuators- Components and typical circuit, advantages and disadvantages; Pneumatic Actuators- Components and typical circuit, advantages and disadvantages.
 English support: definition/meaning.
 Technical terms English-
 English-

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

Select suitable sensors and actuators for robotic systems.

M2.01 Remembering	List the different types of sensors used in robotics, such as touch, force, proximity, and vision sensors.	3
M2.02 Understanding	Explain the difference between internal and external sensors, and describe vision sensor elements.	4
M2.03 Understanding	Describe the classification, working, and advantages/disadvantages of electric, hydraulic, and pneumatic actuators.	4
M2.04 Applying	Apply knowledge to select suitable sensors and actuators for robotic applications.	3
	Series Test - I	1

Contents:

Sensors and Actuators

Sensor classification- touch, force, proximity, vision sensors.

Internal sensors-Position sensors, velocity sensors, acceleration sensors, Force sensors;

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Identify the right robotic configuration and gripper for specific tasks.

This module is presented from the official syllabus scope for Basics of Robotics. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Identify the right robotic configuration and gripper for specific tasks.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

3 Remembering (PPP, RPP, RRP, RRR) and their features. Describe SCARA and PUMA robots and their

3 Remembering (PPP, RPP, RRP, RRR) and their features. Describe SCARA and PUMA robots and their is an official topic in Module 3 of Basics of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Remembering (PPP, RPP, RRP, RRR) and their features. Describe SCARA and PUMA robots and their using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 remembering (ppp, rpp, rrp, rrr) and their features. describe scara and puma robots and their should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Remembering (PPP, RPP, RRP, RRR) and their features. Describe SCARA and PUMA robots and their means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-ഓ 3 Remembering (PPP, RPP, RRP, RRR) and their features. Describe SCARA and PUMA robots and their ഓരോന്നിന്റെയും ഓരോന്നിന്റെയും definition/meaning ഓരോന്നിന്റെയും. ഓരോന്നിന്റെയും ഓരോന്നിന്റെയും steps, application ഓരോന്നിന്റെയും ഓരോന്നിന്റെയും. Technical terms English-ഓ ഓരോന്നിന്റെയും.

3 Understanding uses. Classify types of end effectors such as

3 Understanding uses. Classify types of end effectors such as is an official topic in Module 3 of Basics of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding uses. Classify types of end effectors such as using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding uses. classify types of end effectors such as should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding uses. Classify types of end effectors such as means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 3-3 Understanding uses. Classify types of end effectors such as ഓരോന്നിനും definition/meaning ഓരോന്നിനും. ഓരോന്നും ഓരോന്നും, steps, application ഓരോന്നും ഓരോന്നിനും. Technical terms English-ഓരോന്നും ഓരോന്നിനും.

4 Understanding grippers, tools, and their design considerations. Choose appropriate configurations and

4 Understanding grippers, tools, and their design considerations. Choose appropriate configurations and is an official topic in Module 3 of Basics of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding grippers, tools, and their design considerations. Choose appropriate configurations and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding grippers, tools, and their design considerations. choose appropriate configurations and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding grippers, tools, and their design considerations. Choose appropriate configurations and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-4 Understanding grippers, tools, and their design considerations. Choose appropriate configurations and definition/meaning. steps, application. Technical terms English-

4 Applying grippers for specific applications. Robotic configurations and end effectors Robot configurations-PPP, RPP, RRP, RRR; features of SCARA, PUMA Robots; Classification of robots based on motion control methods and drive technologies; 3R concurrent wrist; Classification of End effectors - mechanical grippers, special tools, Magnetic grippers, Vacuum grippers, adhesive grippers, Active and passive grippers, selection and design considerations of grippers in robot. Determine the most suitable robotic configuration and gripper for a

4 Applying grippers for specific applications. Robotic configurations and end effectors Robot configurations-PPP, RPP, RRP, RRR; features of SCARA, PUMA Robots; Classification of robots based on motion control methods and drive technologies; 3R concurrent wrist; Classification of End effectors - mechanical grippers, special tools, Magnetic grippers, Vacuum grippers, adhesive grippers, Active and passive grippers, selection and design considerations of grippers in robot. Determine the most suitable robotic configuration and gripper for a is an official topic in Module 3 of Basics of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Applying grippers for specific applications. Robotic configurations and end effectors Robot configurations-PPP, RPP, RRP, RRR; features of SCARA, PUMA Robots; Classification of robots based on motion control methods and drive technologies; 3R concurrent wrist; Classification of End effectors - mechanical grippers, special tools, Magnetic grippers, Vacuum grippers, adhesive grippers, Active and passive grippers, selection and design considerations of grippers in robot. Determine the most suitable robotic configuration and gripper for a using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 applying grippers for specific applications. robotic configurations and end effectors robot configurations-ppp, rpp, rrp, rrr; features of scara, puma robots; classification of robots based on motion control methods and drive technologies; 3r concurrent wrist; classification of end effectors - mechanical grippers, special tools, magnetic grippers, vacuum grippers, adhesive grippers, active and passive grippers, selection and design considerations of grippers in robot. determine the most suitable robotic configuration and gripper for a should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Applying grippers for specific applications. Robotic configurations and end effectors Robot configurations-PPP, RPP, RRP, RRR; features of SCARA, PUMA Robots; Classification of robots based on motion control methods and drive technologies; 3R concurrent wrist; Classification of End effectors - mechanical grippers, special tools, Magnetic grippers, Vacuum grippers, adhesive grippers, Active and passive grippers, selection and design considerations of grippers in robot. Determine the most suitable robotic configuration and gripper for a means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- 4 Applying grippers for specific applications. Robotic configurations and end effectors Robot configurations-PPP, RPP, RRP, RRR; features of SCARA, PUMA Robots; Classification of robots based on motion control methods and drive technologies; 3R concurrent wrist; Classification of End effectors - mechanical grippers, special tools, Magnetic grippers, Vacuum grippers, adhesive grippers, Active and passive grippers, selection and design considerations of grippers in robot. Determine the most suitable robotic configuration and gripper for a definition/meaning. steps, application. Technical terms English- .

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

Identify the right robotic configuration and gripper for specific tasks.

	Recall the different robotic configurations		
M3.01		3	
Remembering			
	(PPP, RPP, RRP, RRR) and their features.		
	Describe SCARA and PUMA robots and their		
M3.02		3	
Understanding			
	uses.		
	Classify types of end effectors such as		
M3.03		4	
Understanding			
	grippers, tools, and their design considerations.		
	Choose appropriate configurations and		
M3.04		4	Applying
	grippers for specific applications.		
Contents:			
Robotic configurations and end effectors			
Robot configurations-PPP, RPP, RRP, RRR; features of SCARA, PUMA Robots;			
Classification of robots based on motion control methods and drive technologies;			
3R			
concurrent wrist; Classification of End effectors - mechanical grippers, special			
tools,			
Magnetic grippers, Vacuum grippers, adhesive grippers, Active and passive			
grippers,			
selection and design considerations of grippers in robot.			

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

specific application.

This module is presented from the official syllabus scope for Basics of Robotics. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: specific application.
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

3D space and their composition through 3 Understanding sequential applications of rotations. Explain the representation of rotations using

3D space and their composition through 3 Understanding sequential applications of rotations. Explain the representation of rotations using is an official topic in Module 4 of Basics of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3D space and their composition through 3 Understanding sequential applications of rotations. Explain the representation of rotations using using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3d space and their composition through 3 understanding sequential applications of rotations. explain the representation of rotations using should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3D space and their composition through 3 Understanding sequential applications of rotations. Explain the representation of rotations using means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a

diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-3D space and their composition through 3 Understanding sequential applications of rotations. Explain the representation of rotations using definition/meaning. steps, application. Technical terms English-.

3 Understanding matrices and their properties. Understand the concept of homogeneous

3 Understanding matrices and their properties. Understand the concept of homogeneous is an official topic in Module 4 of Basics of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding matrices and their properties. Understand the concept of homogeneous using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding matrices and their properties. understand the concept of homogeneous should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding matrices and their properties. Understand the concept of homogeneous means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-3 Understanding matrices and their properties. Understand the concept of homogeneous $ax = b$ definition/meaning $ax = b$. $ax = b$ $ax = b$ $ax = b$, steps, application $ax = b$ $ax = b$. Technical terms English- $ax = b$ $ax = b$.

coordinates and transformations for unified 4 Understanding representation of positions and orientations. Apply homogeneous transformation matrices

coordinates and transformations for unified 4 Understanding representation of positions and orientations. Apply homogeneous transformation matrices is an official topic in Module 4 of Basics of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify coordinates and transformations for unified 4 Understanding representation of positions and orientations. Apply homogeneous transformation matrices using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, coordinates and transformations for unified 4 understanding representation of positions and orientations. apply homogeneous transformation matrices should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What coordinates and transformations for unified 4 Understanding representation of positions and orientations. Apply homogeneous transformation matrices means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a

diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- 3D coordinates and transformations for unified 4 Understanding representation of positions and orientations. Apply homogeneous transformation matrices 4×4 definition/meaning 4×4 matrices. 4×4 matrices 4×4 matrices, steps, application 4×4 matrices 4×4 matrices. Technical terms English- 4×4 matrices 4×4 matrices.

to describe translations, rotations, and scaling 4 Applying in robotic systems. Use forward and inverse kinematics to

to describe translations, rotations, and scaling 4 Applying in robotic systems. Use forward and inverse kinematics to is an official topic in Module 4 of Basics of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify to describe translations, rotations, and scaling 4 Applying in robotic systems. Use forward and inverse kinematics to using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, to describe translations, rotations, and scaling 4 applying in robotic systems. use forward and inverse kinematics to should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What to describe translations, rotations, and scaling 4 Applying in robotic systems. Use forward and inverse kinematics to means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- to describe translations, rotations, and scaling 4 Applying in robotic systems. Use forward and inverse kinematics to definition/meaning . steps, application . Technical terms English- .

determine the positions and orientations of 2 Applying robotic arms. Kinematics and Motion Planning Understanding basic rotational motions in 2D and 3D space, Composition of rotations: sequential application of rotations around different axes, Representation of rotations using matrices and their properties, Homogeneous Coordinates and Transformations, Representation of positions and orientations of robotic components in a unified framework. Use of homogeneous transformation matrices to describe translations, rotations, and scaling in a single matrix. Applications in robotic arm manipulation, such as forward and inverse kinematics. Text /Reference: T/R BookTitle/Author T1 S K Saha ,Introduction to Robotics,, Mc Graw Hill Eduaction Robert. J. Schilling , “Fundamentals of robotics - Analysis and control”, T2 Prentice Hall of India 1996. R K Mittal and I J Nagrath, “Robotics and Control”, Tata McGraw Hill, New T3 Delhi, 2003. John. J. Craig , Introduction to Robotics (Mechanics and control), Pearson R1 Education Asia 2002. Ashitava Ghosal, “Robotics-Fundamental concepts and analysis”, Oxford R2 University press Online resources: Sl.No Website Link

https://www.robotics.org 1 https://www.ros.org/ 2
https://www.allaboutcircuits.com/textbook/direct-current/chpt- 3 9/overview-of-sensors/ https://www.industrial-robotics.org/ 4
https://ocw.mit.edu/courses/mechanical-engineering/2-12-introduction- 5 to-robotics-fall-2007/

determine the positions and orientations of 2 Applying robotic arms. Kinematics and Motion Planning Understanding basic rotational motions in 2D and 3D space, Composition of rotations: sequential application of rotations around different axes, Representation of rotations using matrices and their properties, Homogeneous Coordinates and Transformations, Representation of positions and orientations of robotic components in a unified framework. Use of homogeneous transformation matrices to describe translations, rotations, and scaling in a single matrix. Applications in robotic arm manipulation, such as forward and inverse kinematics. Text /Reference: T/R BookTitle/Author T1 S K Saha ,Introduction to Robotics,, Mc Graw Hill Eduaction Robert. J. Schilling , “Fundamentals of robotics – Analysis and control”, T2 Prentice Hall of India 1996. R K Mittal and I J Nagrath, “Robotics and Control”, Tata McGraw Hill, New T3 Delhi, 2003. John. J. Craig , Introduction to Robotics (Mechanics and control), Pearson R1 Education Asia 2002. Ashitava Ghosal, “Robotics-Fundamental concepts and analysis”, Oxford R2 University press Online resources: Sl.No Website Link https://www.robotics.org 1 https://www.ros.org/ 2 https://www.allaboutcircuits.com/textbook/direct-current/chpt- 3 9/overview-of-sensors/ https://www.industrial-robotics.org/ 4 https://ocw.mit.edu/courses/mechanical-engineering/2-12-introduction- 5 to-robotics-fall-2007/ is an official topic in Module 4 of Basics of Robotics. Study the

terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify determine the positions and orientations of 2 Applying robotic arms. Kinematics and Motion Planning Understanding basic rotational motions in 2D and 3D space, Composition of rotations: sequential application of rotations around different axes, Representation of rotations using matrices and their properties, Homogeneous Coordinates and Transformations, Representation of positions and orientations of robotic components in a unified framework. Use of homogeneous transformation matrices to describe translations, rotations, and scaling in a single matrix. Applications in robotic arm manipulation, such as forward and inverse kinematics. Text /Reference: T/R BookTitle/Author T1 S K Saha ,Introduction to Robotics,, Mc Graw Hill Eduaction Robert. J. Schilling , “Fundamentals of robotics – Analysis and control”, T2 Prentice Hall of India 1996. R K Mittal and I J Nagrath, “Robotics and Control”, Tata McGraw Hill, New T3 Delhi, 2003. John. J. Craig , Introduction to Robotics (Mechanics and control), Pearson R1 Education Asia 2002. Ashitava Ghosal, “Robotics-Fundamental concepts and analysis”, Oxford R2 University press Online resources: Sl.No Website Link
<https://www.robotics.org> 1 <https://www.ros.org/> 2
[https://www.allaboutcircuits.com/textbook/direct-current/chpt- 3 9/overview-of-sensors/](https://www.allaboutcircuits.com/textbook/direct-current/chpt-39/overview-of-sensors/) <https://www.industrial-robotics.org/> 4
[https://ocw.mit.edu/courses/mechanical-engineering/2-12-introduction- 5 to-robotics-fall-2007/](https://ocw.mit.edu/courses/mechanical-engineering/2-12-introduction-5-to-robotics-fall-2007/) using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, determine the positions and orientations of 2 applying robotic arms. kinematics and motion planning understanding basic rotational motions in 2d and 3d space, composition of rotations: sequential application of rotations around different axes, representation of rotations using matrices and their properties, homogeneous coordinates and transformations, representation of positions and orientations of robotic components in a unified framework. use of homogeneous transformation matrices to

describe translations, rotations, and scaling in a single matrix. applications in robotic arm manipulation, such as forward and inverse kinematics. text /reference: t/r booktitle/author t1 s k saha ,introduction to robotics,, mc graw hill eduaction robert. j. schilling , “fundamentals of robotics – analysis and control”, t2 prentice hall of india 1996. r k mittal and i j nagrath, “robotics and control”, tata mcgraw hill, new t3 delhi, 2003. john. j. craig , introduction to robotics (mechanics and control), pearson r1 education asia 2002. ashitava ghosal, “robotics-fundamental concepts and analysis”, oxford r2 university press online resources: sl.no website link <https://www.robotics.org> 1 <https://www.ros.org/> 2 [https://www.allaboutcircuits.com/textbook/direct-current/chpt- 3 9/overview-of-sensors/](https://www.allaboutcircuits.com/textbook/direct-current/chpt-3/9/overview-of-sensors/) <https://www.industrial-robotics.org/> 4 [https://ocw.mit.edu/courses/mechanical-engineering/2-12-introduction- 5 to-robotics-fall-2007/](https://ocw.mit.edu/courses/mechanical-engineering/2-12-introduction-5-to-robotics-fall-2007/) should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What determine the positions and orientations of 2 Applying robotic arms. Kinematics and Motion Planning Understanding basic rotational motions in 2D and 3D space, Composition of rotations: sequential application of rotations around different axes, Representation of rotations using matrices and their properties, Homogeneous Coordinates and Transformations, Representation of positions and orientations of robotic components in a unified framework. Use of homogeneous transformation matrices to describe translations, rotations, and scaling in a single matrix. Applications in robotic arm manipulation, such as forward and inverse kinematics. Text /Reference: T/R BookTitle/Author T1 S K Saha ,Introduction to Robotics,, Mc Graw Hill Eduaction Robert. J. Schilling , “Fundamentals of robotics – Analysis and control”, T2 Prentice Hall of India 1996. R K Mittal and I J Nagrath, “Robotics and Control”, Tata McGraw Hill, New T3 Delhi, 2003. John. J. Craig , Introduction to Robotics (Mechanics and control), Pearson R1 Education Asia 2002. Ashitava Ghosal, “Robotics-Fundamental concepts and analysis”, Oxford R2 University press Online resources: Sl.No Website Link https://www.robotics.org 1 https://www.ros.org/ 2 https://www.allaboutcircuits.com/textbook/direct-current/chpt- 3 9/overview-of-sensors/ https://www.industrial-robotics.org/ 4 https://ocw.mit.edu/courses/mechanical-engineering/2-12-introduction- 5 to-robotics-fall-2007/ means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- determine the positions and orientations of 2 Applying robotic arms. Kinematics and Motion Planning Understanding basic rotational motions in 2D and 3D space, Composition of rotations: sequential application of rotations around different axes, Representation of rotations using

matrices and their properties, Homogeneous Coordinates and Transformations, Representation of positions and orientations of robotic components in a unified framework. Use of homogeneous transformation matrices to describe translations, rotations, and scaling in a single matrix. Applications in robotic arm manipulation, such as forward and inverse kinematics. Text /Reference: T/R BookTitle/Author T1 S K Saha ,Introduction to Robotics,, Mc Graw Hill Education Robert. J. Schilling , “Fundamentals of robotics – Analysis and control”, T2 Prentice Hall of India 1996. R K Mittal and I J Nagrath, “Robotics and Control”, Tata McGraw Hill, New T3 Delhi, 2003. John. J. Craig , Introduction to Robotics (Mechanics and control), Pearson R1 Education Asia 2002. Ashitava Ghosal, “Robotics-Fundamental concepts and analysis”, Oxford R2 University press Online resources: Sl.No Website Link
<https://www.robotics.org> 1 <https://www.ros.org/> 2
[https://www.allaboutcircuits.com/textbook/direct-current/chpt- 3 9/overview-of-sensors/](https://www.allaboutcircuits.com/textbook/direct-current/chpt-39/overview-of-sensors/) <https://www.industrial-robotics.org/> 4
[https://ocw.mit.edu/courses/mechanical-engineering/2-12-introduction- 5 to-robotics-fall-2007/](https://ocw.mit.edu/courses/mechanical-engineering/2-12-introduction-5-to-robotics-fall-2007/) [definition/meaning](#) [steps](#), application [Technical terms](#) English-[English](#) [English](#).

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

specific application.

M4.01	Describe basic rotational motions in 2D and 3D space and their composition through	3
Understanding	sequential applications of rotations.	
M4.02	Explain the representation of rotations using	3
Understanding	matrices and their properties.	
M4.03	Understand the concept of homogeneous coordinates and transformations for unified	4
Understanding	representation of positions and orientations.	
M4.04	Apply homogeneous transformation matrices to describe translations, rotations, and scaling	4
Applying	in robotic systems.	
M4.05	Use forward and inverse kinematics to determine the positions and orientations of	2
Applying	robotic arms.	
	Series Test - II	1

Contents:
Kinematics and Motion Planning
Understanding basic rotational motions in 2D and 3D space, Composition of rotations:

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

Definition → Principle → Components/steps → Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure → Observation → Result → Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain 3 Remembering types of robots (manipulators, mobile, aerial). Describe the anatomy of a robotic manipulator,. (3 marks)

Answer: Begin with the standard definition or identity of *3 Remembering types of robots (manipulators, mobile, aerial)*. Describe the anatomy of a robotic manipulator,. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain including links, joints, actuators, and 5 Understanding controllers. Explain kinematic chains, degrees of freedom,. (3 marks)

Answer: Begin with the standard definition or identity of *including links, joints, actuators, and 5 Understanding controllers. Explain kinematic chains, degrees of freedom,.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain 4 Understanding and operational considerations for robots. List applications of robots in various fields like. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding and operational considerations for robots. List applications of robots in various fields like.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain 2 Remembering medical, industrial, mining, and space. Definition, Anatomy, and Applications of Robots Definitions- Robots, Robotics; Types of Robots- Manipulators, Mobile Robots-wheeled & Legged Robots, Aerial Robots; Anatomy of a robotic manipulator-links, joints, actuators, sensors, controller; open kinematic vs closed kinematic chain; degrees of freedom; Robot considerations for an application- number of axes, work volume, capacity & speed, stroke & reach, Repeatability, Precision and Accuracy, Operating environment, point to point control or continuous path control., Robot Applications- medical, mining, space, defense, security,

domestic, entertainment, Industrial Applications-Material handling, welding, Spray painting, Machining.. (3 marks)

Answer: Begin with the standard definition or identity of 2 Remembering medical, industrial, mining, and space. Definition, Anatomy, and Applications of Robots Definitions- Robots, Robotics; Types of Robots- Manipulators, Mobile Robots-wheeled & Legged Robots, Aerial Robots; Anatomy of a robotic manipulator-links, joints, actuators, sensors, controller; open kinematic vs closed kinematic chain; degrees of freedom; Robot considerations for an application- number of axes, work volume, capacity & speed, stroke & reach, Repeatability, Precision and Accuracy, Operating environment, point to point control or continuous path control., Robot Applications- medical, mining, space, defense, security, domestic, entertainment, Industrial Applications-Material handling, welding, Spray painting, Machining.. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: ഏകദേശം definition, പ്രധാനപ്പെട്ട principle/steps, ഉദാഹരണം application നൽകി വിശദീകരിക്കുക.

Module 2

Define or explain robotics, such as touch, force, proximity, and 3 Remembering vision sensors. Explain the difference between internal and. (3 marks)

Answer: Begin with the standard definition or identity of robotics, such as touch, force, proximity, and 3 Remembering vision sensors. Explain the difference between internal and. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: ഏകദേശം definition, പ്രധാനപ്പെട്ട principle/steps, ഉദാഹരണം application നൽകി വിശദീകരിക്കുക.

Define or explain external sensors, and describe vision sensor 4 Understanding elements. Describe the classification, working, and. (3 marks)

Answer: Begin with the standard definition or identity of external sensors, and describe vision sensor 4 Understanding elements. Describe the classification, working, and. State

the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain advantages/disadvantages of electric, 4 Understanding hydraulic, and pneumatic actuators. Apply knowledge to select suitable sensors and. (3 marks)

Answer: Begin with the standard definition or identity of *advantages/disadvantages of electric, 4 Understanding hydraulic, and pneumatic actuators. Apply knowledge to select suitable sensors and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 3 Applying actuators for robotic applications. Sensors and Actuators Sensor classification- touch, force, proximity, vision sensors. Internal sensors-Position sensors, velocity sensors, acceleration sensors, Force sensors; External sensors-contact type, noncontact type; Vision - Elements of vision sensor, image acquisition, image processing; Selection of sensors. Actuators for robots- classification-Electric, Hydraulic, Pneumatic actuators; their advantages and disadvantages; Electric actuators- Stepper motors, DC motors, DC servo motors and their drivers, AC motors, Linear actuators, selection of motors; Hydraulic actuators- Components and typical circuit, advantages and disadvantages; Pneumatic Actuators- Components and typical circuit, advantages and disadvantages.. (3 marks)

Answer: Begin with the standard definition or identity of *3 Applying actuators for robotic applications. Sensors and Actuators Sensor classification- touch, force, proximity, vision sensors. Internal sensors-Position sensors, velocity sensors, acceleration sensors, Force sensors; External sensors-contact type, noncontact type; Vision - Elements of vision sensor, image acquisition, image processing; Selection of sensors. Actuators for robots- classification-Electric, Hydraulic, Pneumatic actuators; their advantages and disadvantages; Electric actuators- Stepper motors, DC motors, DC servo motors and*

their drivers, AC motors, Linear actuators, selection of motors; Hydraulic actuators- Components and typical circuit, advantages and disadvantages; Pneumatic Actuators- Components and typical circuit, advantages and disadvantages. . State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Module 3

Define or explain 3 Remembering (PPP, RPP, RRP, RRR) and their features. Describe SCARA and PUMA robots and their. (3 marks)

Answer: Begin with the standard definition or identity of 3 Remembering (PPP, RPP, RRP, RRR) and their features. Describe SCARA and PUMA robots and their. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain 3 Understanding uses. Classify types of end effectors such as. (3 marks)

Answer: Begin with the standard definition or identity of 3 Understanding uses. Classify types of end effectors such as. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain 4 Understanding grippers, tools, and their design considerations. Choose appropriate configurations and. (3 marks)

Answer: Begin with the standard definition or identity of 4 *Understanding grippers, tools, and their design considerations. Choose appropriate configurations and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 4 Applying grippers for specific applications. Robotic configurations and end effectors Robot configurations-PPP, RPP, RRP, RRR; features of SCARA, PUMA Robots; Classification of robots based on motion control methods and drive technologies; 3R concurrent wrist; Classification of End effectors - mechanical grippers, special tools, Magnetic grippers, Vacuum grippers, adhesive grippers, Active and passive grippers, selection and design considerations of grippers in robot. Determine the most suitable robotic configuration and gripper for a. (3 marks)

Answer: Begin with the standard definition or identity of 4 *Applying grippers for specific applications. Robotic configurations and end effectors Robot configurations-PPP, RPP, RRP, RRR; features of SCARA, PUMA Robots; Classification of robots based on motion control methods and drive technologies; 3R concurrent wrist; Classification of End effectors - mechanical grippers, special tools, Magnetic grippers, Vacuum grippers, adhesive grippers, Active and passive grippers, selection and design considerations of grippers in robot. Determine the most suitable robotic configuration and gripper for a.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 4

Define or explain 3D space and their composition through 3 Understanding sequential applications of rotations. Explain the representation of rotations using. (3 marks)

Answer: Begin with the standard definition or identity of *3D space and their composition through 3 Understanding sequential applications of rotations. Explain the representation of rotations using*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 3 Understanding matrices and their properties. Understand the concept of homogeneous. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding matrices and their properties. Understand the concept of homogeneous*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain coordinates and transformations for unified 4 Understanding representation of positions and orientations. Apply homogeneous transformation matrices. (3 marks)

Answer: Begin with the standard definition or identity of *coordinates and transformations for unified 4 Understanding representation of positions and orientations. Apply homogeneous transformation matrices*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain to describe translations, rotations, and scaling 4 Applying in robotic systems. Use forward and inverse kinematics to. (3 marks)

Answer: Begin with the standard definition or identity of *to describe translations, rotations, and scaling* 4 Applying in robotic systems. Use forward and inverse kinematics to. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **3 Remembering types of robots (manipulators, mobile, aerial). Describe the anatomy of a robotic manipulator.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **robotics, such as touch, force, proximity, and 3 Remembering vision sensors. Explain the difference between internal and.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **3 Remembering (PPP, RPP, RRP, RRR) and their features. Describe SCARA and PUMA robots and their.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **3D space and their composition through 3 Understanding sequential applications of rotations. Explain the representation of rotations using.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.

- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link <https://www.robotics.org> <https://www.ros.org/>
- <https://www.allaboutcircuits.com/textbook/direct-current/chpt- /overview-of-sensors/>
<https://www.industrial-robotics.org/>
- <https://ocw.mit.edu/courses/mechanical-engineering/2-12-introduction- to-robotics-fall-2007/>

IN-PAGE SEARCH

Search this lesson

Prepared from the supplied official SITTTR syllabus PDF for Course 6492B. Verify institutional updates against the current official curriculum.